Chapter 9: Maxwell's Equations

The following subjects will be discussed

- Faraday's Law of Induction.
- Maxwell's Equations.

Faraday's Law

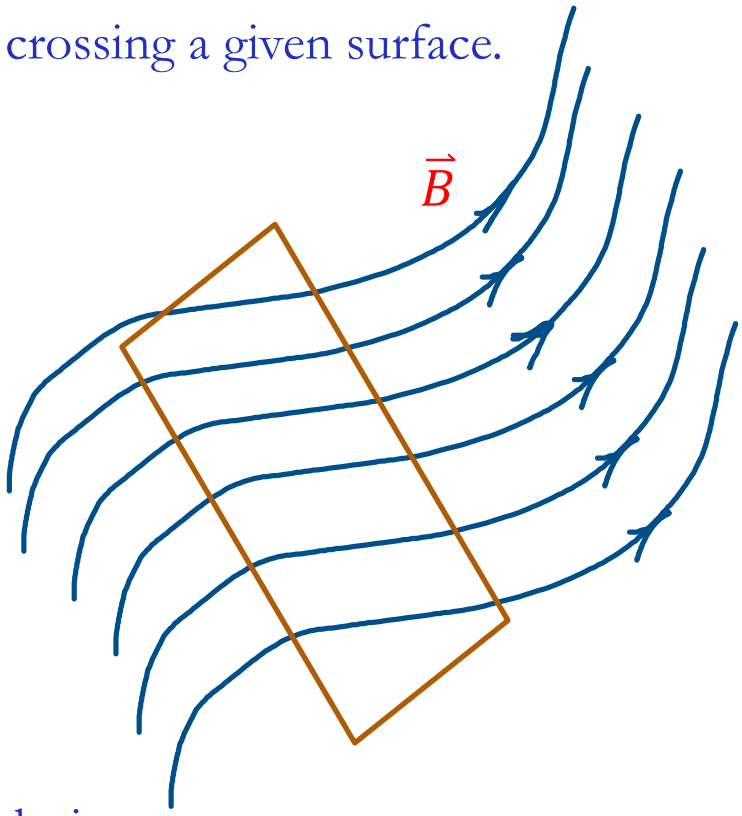


Faraday's Law

Magnetic flux is the number of magnetic field lines crossing a given surface.

Flux equals the surface integral of the field over the given surface.

$$\phi = \int \vec{B} \cdot \overrightarrow{dA}$$



If there is a change in the linkage in the wire (circuit) with respect to time, in any way, an emf will be generated within the wire (circuit)

$$Emf = -\frac{d\phi}{dt}$$

Faraday's Law Of induction

Ways to change linkage in time:

(1) B changing with time.

(2) Area changes with time

(3) Circuit is moving

(4) Combinations of the above.

Induced Electric Field

Let's find a direct relation between the change in magnetic field and the corresponding induced electric field.

For a time-varying magnetic field, we can write

$$Emf = -\frac{d\phi}{dt}$$

$$Emf = -\frac{d\phi}{dt} = -\int \frac{d\vec{B}}{dt} \cdot \overrightarrow{dA}$$

We know that

$$V = -\int \vec{E} \cdot \overrightarrow{dl} \quad \Rightarrow \quad Emf = \oint \vec{E} \cdot \overrightarrow{dl}$$

So, we have

$$\oint \vec{E} \cdot \overrightarrow{dl} = -\frac{d\phi}{dt} = -\int \frac{d\vec{B}}{dt} \cdot \overrightarrow{dA}$$

Note:
The electric field is generated even in the absence of a circuit!

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

Case (1): Transformational EMF

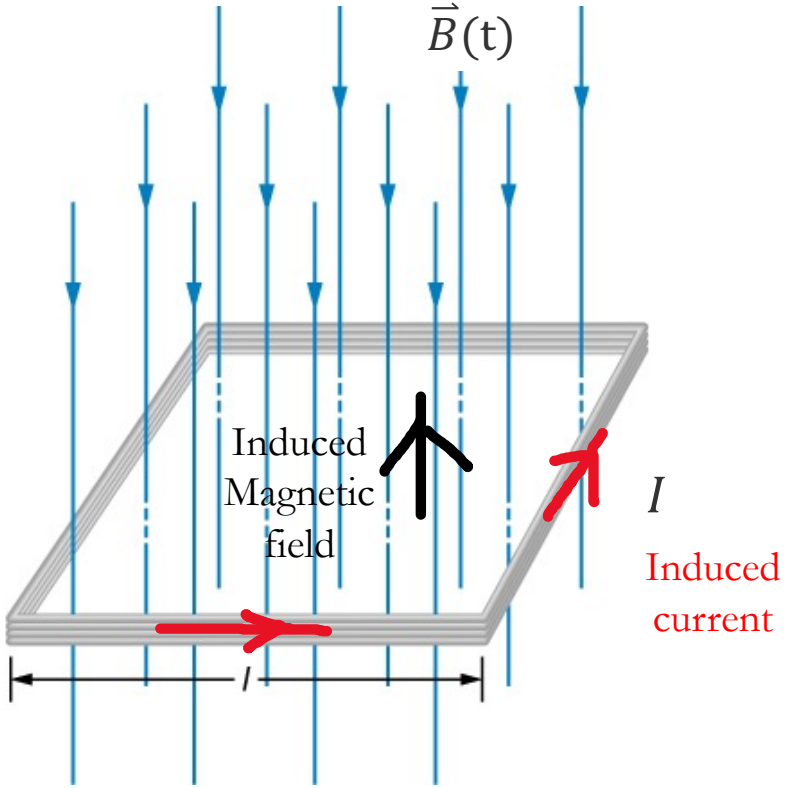
A conducting wire (circuit) with N turns is stationary inside a region of uniform and time-varying magnetic field. There is a change in flux linking the circuit with time.

$$\phi(t) = N \int \vec{B}(t) \cdot \overrightarrow{dA}$$

$$Emf = -\frac{d\phi\,(t)}{dt}$$

$$Emf = -N \int \frac{d\vec{B}(t)}{dt} \cdot \overrightarrow{dA}$$

The magnitude of the EMF can be found using the above equation (without the minus sign). The polarity of the EMF (or direction of induced current in the circuit) can be found using Lenz's law.



Case (2): Motional EMF

A conducting rod is moved inside a region of constant magnetic field.

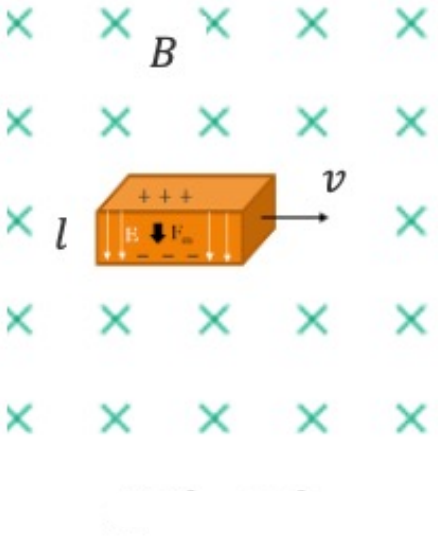
Change in flux is not involved here. However, magnetic and electric forces are applied to the charges within the conductor

At equilibrium,

$$\vec{F}_m = \vec{F}_e$$

$$q\vec{E} = q(\vec{v}\times\vec{B})$$

$$Emf = \int \vec{E}\cdot\overrightarrow{dl} = \int (\vec{v}\times\vec{B})\cdot\overrightarrow{dl}$$



For constant velocity and uniform magnetic field,

$$Emf = vBl$$

Case (2): Motional EMF

A conducting rod is moved inside a region of constant magnetic field.

Change in flux is not involved here. However, magnetic and electric forces are applied to the charges within the conductor

For constant angular velocity and uniform magnetic field,

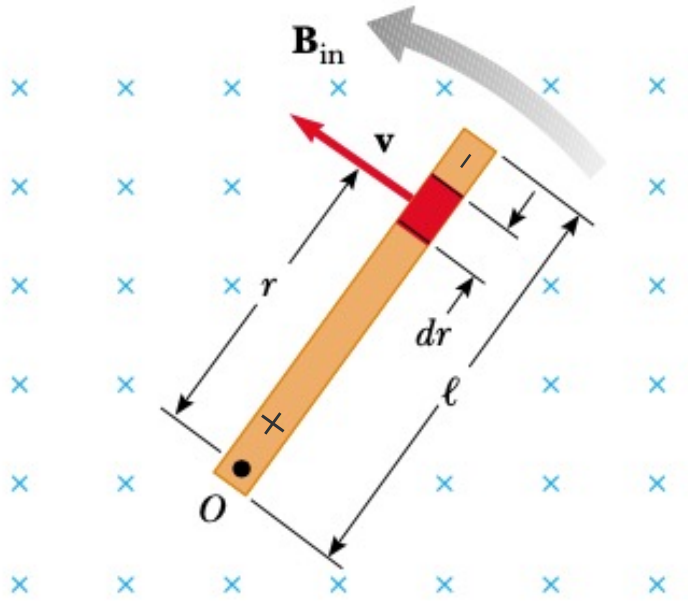
$$Emf = vBl$$

$$d(Emf) = vB(dr)$$

$$v = \omega r$$

$$Emf = \int_0^l \omega B\, r\, dr$$

$$Emf = \frac{1}{2}\omega B l^2$$



Case (3): Transformational and Motional EMF

A part of the circuit is moved inside a region of time-varying magnetic field.

This case can be solved by adding the change in flux linking the circuit, and motional consideration.

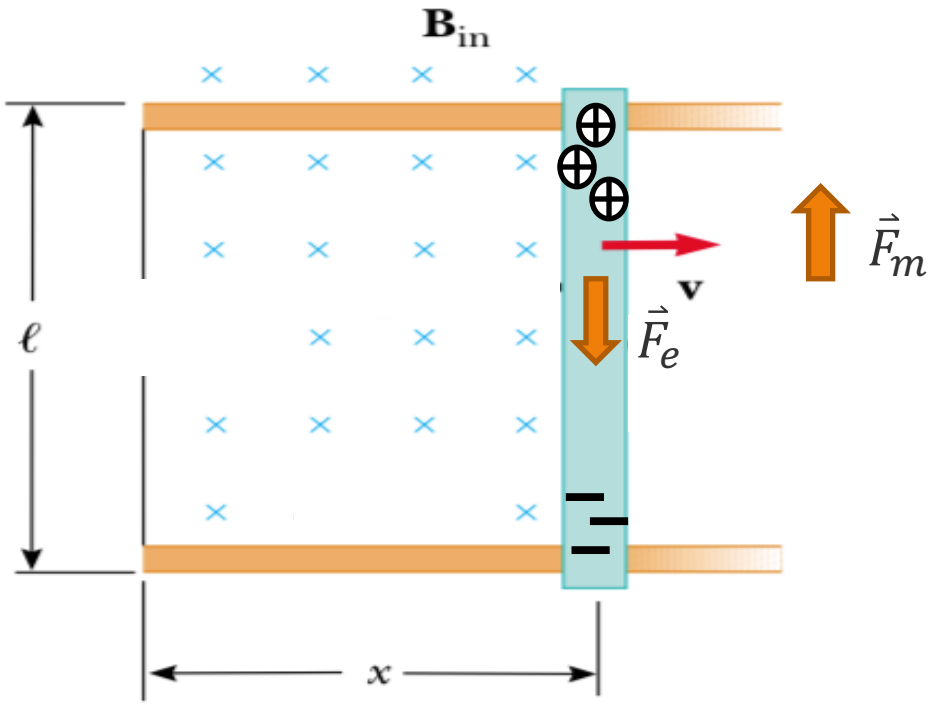
$$Emf = -\int \frac{d\vec{B}(t)}{dt} \cdot \overrightarrow{dA} + \int (\vec{v} \times \vec{B}) \cdot \overrightarrow{dl}$$

In this case

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} + \vec{\nabla} \times (\vec{v} \times \vec{B})$$

Note that it is also possible to write the total time-varying flux, taking into account the moving part, and then using

$$Emf = -\frac{d\phi\,(t)}{dt}$$



Current direction is such that it opposes the change in linkage!

Lenz's Law

Q & A

Send me an email if you have any questions.